

PRAJWAL KUMAR K

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#) | [AtCoder](#)

EDUCATION

Indian Institute of Information Technology, Kottayam

B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0

Kottayam, Kerala

Aug 2024 – May 2028 (expected)

TECHNICAL SKILLS

Programming: C++, Rust, Python, TypeScript, JavaScript, C, Java, SQL, Solidity, MATLAB

AI/ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

Research and engineering: Jupyter, Simulink, FastAPI, Next.js

Research interests: Algorithms and systems, game theory, machine learning and NLP, AI safety

Languages: English (Fluent), German (Business Proficiency), Hindi, Kannada

EXPERIENCE

Full-Stack Web Developer Intern

Dec 2025 – Mar 2026

CIO Mogul (Magazine)

Remote

- Shipped three magazine websites—[CIO Mogul](#), [CIO Mogul India](#), and [CIO Healthcare](#)—for editorial publishing and digital issues.
- Developed Next.js/React editorial dashboards and Supabase/PostgreSQL content workflows, including PDF page navigation, cover extraction, and A3-spread splitting.

Technical Lead, Openverse (Open Source Club)

May 2025 – Jul 2026

IIT Kottayam

Kottayam, Kerala

- Contributed sorting visualizations to Algorithmia; reviewed pull requests and coordinated Openverse technical work.

Coordinator, Coders Club

Jan 2025 – Present

IIT Kottayam

Kottayam, Kerala

- Curate problem sets, configure Codeforces Gym rounds, and organize campus programming contests.

PROJECTS

PokeForge | *Python, PyTorch, Gen 9 OU*

[Code](#)

- An agent-evaluation testbed for competitive Generation 9 OU Pokémon battles (in progress).
- Integrated pinned FoulPlay/Metamon baselines; 61 offline tests cover selected action, state, and damage contracts while simulator conformance remains under evaluation.

Bluff | *Imperfect-information games, opponent modeling*

[Code](#)

- A repeated-play imperfect-information card game for studying bluff detection and opponent adaptation.
- Implemented Bayesian opponent profiles and a learned public-information bluff detector; ablation lowered draw-aware score by 4.9–5.8 percentage points across eight synthetic profiles.

Drishti-XAI | *Retinal imaging, MATLAB, Python*

- An explainable retinal-screening prototype evaluated for generalization across diabetic-retinopathy image datasets.
- Built the MATLAB/Python prototype and evaluation pipeline; at a fixed model and threshold, sensitivity/specificity were 0.908/0.877 on APTOS (n=733) and 0.363/0.865 on Messidor-2 (AUC 0.666).

SiliconSpire | *C++, Rust, Metaheuristics*

[C++](#) | [Rust](#)

- A Quadratic Assignment Problem solver that minimizes weighted flow-distance cost for facility placement.
- Combined Grey Wolf Optimization with swap-based Tabu Search; across three small QAPLIB instances and 90 seeds per implementation, Rust matched published optima in 77 runs and C++ in 34.

BattleCP | *Rust, Axum, Tokio, WebSockets*

[Live](#) | [Code](#)

- A multiplayer Battleship game where solving Codeforces problems earns players turns.
- Implemented a Rust/Axum/Tokio WebSocket backend with Codeforces validation, server-side rules, and reconnect restoration.

ACHIEVEMENTS AND CERTIFICATIONS

Publication: First-listed author and presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025. [Paper](#).

Algorithmia: selected for and presented at OOSC 3.0, IIT Kanpur (2025); team presentation at T4E 2025, IIT Madras.

Hackathon: Best Software Award, IEEE MACE .hack(); 2026, Team Hogorithm (Anchor).

Competitive programming: Codeforces Pupil (peak 1221), CodeChef peak 1463, 400+ problems solved across platforms.

Cyfrin Updraft (2025): Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.